

David Manuel

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with over 10 years of experience designing and scaling high-performance enterprise platforms, cloud-native backend microservices, complex planner interfaces, and AI-enabled routing and navigation workflows. Tech lead with direct experience driving 0→1 initiatives, optimizing end-to-end routing/execution metrics, and leveraging AI tools to accelerate code refactors and unblock key customer deals. Proven track record of direct customer engagement, cross-functional partnership with product and design, and mentoring junior engineers. Seeking a fully remote Senior Software Engineer II / Tech Lead role.

CORE COMPETENCIES

Languages & Core: Python, C++, SQL, TypeScript, JavaScript

Simulation & Systems: Simulation Environments, Autonomous System Software, Robotics Systems, Machine Learning & Computer Vision Integration, Failure Case Analysis

Cloud & DevOps: Google Cloud Platform (GCP), AWS (EKS, ECS, Lambda), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Data & Architectures: Relational & Large-Scale Databases (PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, Redis), REST & GraphQL APIs, Microservices, Distributed Systems

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Architected and delivered end-to-end features across frontend (React, Next.js, TypeScript) and backend microservices (Node.js, Python/FastAPI, Java Spring Boot, .NET Core) powering complex workflow and route execution dashboards.
- Leveraged AI developer tools (Cursor, GitHub Copilot, Claude API, LangChain) to prototype custom customer solutions rapidly, unblocking strategic enterprise deals and accelerating risk-reducing technical refactors.
- Operated as Technical Lead for a 5-person engineering team, establishing architecture standards, leading technical design reviews, and directly mentoring junior engineers to build a strong technical bench.
- Partnered with Product Managers, Designers, and Customer Success in direct customer interviews to gather routing and operational requirements, translating feedback into scalable planner interfaces and API endpoints.
- Optimized routing and data execution latency, cutting median API response times from ~850ms to 380ms (55% improvement) using Redis caching, asynchronous processing, and optimized spatial/vector database indexing across PostgreSQL and Pinecone.
- Implemented key telemetry and performance metrics to track feature adoption and execution efficiency, ensuring engineering deliverables measurably improved core customer outcomes.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Developed full-stack web applications and microservices using React, TypeScript, C#, .NET Core, and Python/FastAPI, driving reliable 0→1 features from initial design to post-startup scaling.
- Reduced p95 backend API latency by 24% (620ms to 470ms) through database query optimization, asynchronous workflow execution, and distributed system tuning.
- Built and secured high-availability RESTful and GraphQL APIs implementing OAuth2, JWT, RBAC, field-level encryption, and strict compliance audit logging.
- Containerized and orchestrated cloud services on AWS (ECS, EKS) and Google Cloud Platform using Docker, Kubernetes, and Terraform with automated CI/CD pipelines.
- Engaged directly with enterprise stakeholders and clinical end-users to refine user interfaces, leading technical discussions and shaping product roadmaps.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Built scalable backend services and API architectures (Java Spring Boot, Node.js, Python) supporting enterprise workflow automation and third-party platform integrations.
- Accelerated average workflow execution time by 50% (12 minutes down to under 6 minutes) through Redis caching, query tuning, and database optimization.
- Automated backend CI/CD pipelines and zero-downtime deployments using Docker, Kubernetes, GitHub Actions, and Azure DevOps.
- Mentored junior engineers on Agile delivery, API design, clean code practices, and system observability.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston